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Human-Centric AI Governance in Digital HRM: A Conceptual Framework for Responsible Digital Business

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Abstract

Artificial intelligence (AI) is increasingly embedded in human resource management (HRM), reshaping recruitment, performance management, workforce analytics, learning, and employee experience. Although these developments create opportunities for efficiency, scalability, and decision support, they also generate governance challenges related to fairness, opacity, privacy, accountability, trust, and employee voice. Existing research has examined responsible AI, digital transformation, and AI-enabled HRM, yet these conversations remain insufficiently integrated. In particular, digital business scholarship has not fully theorized HRM as a strategic governance site within digital transformation. This paper addresses that gap by developing a human-centric conceptual framework for AI governance in digital HRM and positioning it as a foundation for responsible digital business. Drawing on sociotechnical systems theory, governance theory, and stakeholder theory, the framework identifies four interrelated dimensions: governance architecture, human-centric design principles, digital HRM application domains, and responsible digital business outcomes. The paper further develops propositions explaining how these dimensions interact to shape trust, legitimacy, ethical resilience, and sustainable business value. The study contributes to digital business research by positioning digital HRM as a core governance domain within digital transformation, and to HRM scholarship by reframing AI-enabled people management as a sociotechnical and governance-sensitive phenomenon. It also offers practical guidance for organizations seeking to align AI adoption in HRM with responsible digital business outcomes.

Keywords

Artificial intelligence; Digital HRM; AI governance; Responsible digital business; Human-centric AI; Digital transformation

1. Introduction

The rapid diffusion of artificial intelligence (AI) across organizational functions has intensified scholarly and managerial attention to the digitalization of business and its implications for work, management, and value creation. This context is highly relevant to *Digital Business*, which focuses on the digitalization of business, the effects of digital technologies on business models, and implications for the future of work (Digital Business, n.d.). Within this broader transformation, AI-enabled human resource management (HRM) has emerged as a particularly significant domain because it links digital innovation to organizational governance, workforce management, and business legitimacy.

AI is now increasingly embedded in recruitment, selection, performance management, workforce analytics, learning and development, and employee interaction systems. Early research highlighted both the promise and the risks of these technologies, showing that AI can enhance efficiency, scalability, and decision support while also creating concerns related to bias, opacity, and reduced human judgment (Tambe et al., 2019). More recent work confirms that AI-enabled HRM has become an expanding area of inquiry shaped by automation, prediction, personalization, and data-driven people management (Bujold et al., 2024; Úbeda-García et al., 2025). At the same time, broader management scholarship shows that algorithmic management has moved from an edge case to a central concern in organizational research, particularly because it reshapes work allocation, managerial control, and employee experience (Keegan & Meijerink, 2025; Sarala et al., 2025).

The significance of digital HRM extends beyond the HR function itself. Digital transformation research shows that digitalization is not simply the introduction of new tools into existing workflows; it involves the reconfiguration of business models, organizational capabilities, and value creation processes (Ritter & Pedersen, 2020; Vial, 2019). Elia et al. (2024) similarly argue that digital transformation requires alignment among structures, roles, competencies, and organizational enablers. From this perspective, AI in HRM should not be treated as a narrow technical upgrade. Rather, it should be understood as part of a broader digital business transformation that influences how organizations govern work, manage human capital, and sustain legitimacy in data-intensive environments (Elia et al., 2024; Ritter & Pedersen, 2020).

This broader framing is important because organizational enthusiasm for AI often advances faster than governance capability. Research on responsible AI governance shows that although principles such as fairness, transparency, accountability, and oversight are widely discussed, their implementation in organizations remains uneven and fragmented (Papagiannidis et al., 2025). This challenge is reinforced by recent review evidence showing that responsible AI often fails in practice because organizations struggle to move from abstract principles to operational routines, clear accountability, and context-sensitive implementation (Sadek et al., 2025). Recent research in business and management also shows that responsible AI is increasingly being treated as a strategic concern linked not only to risk management but also to innovation, governance, and organizational legitimacy (Dhaigude & Kamath, 2025; Ye et al., 2025). This challenge becomes especially acute in HRM because AI-mediated decisions directly affect access to employment, developmental opportunities, performance evaluations, and workplace experience.

Recent HRM scholarship reinforces this point. Bujold et al. (2024) show that responsible AI in HRM cannot be reduced to technical performance alone, but must also account for ethical, human, and quality-of-work considerations. Kim et al. (2025) similarly argue that algorithmic technologies are reshaping the foundations of people management and require new theorizing about the relationship between HR strategy, managerial judgment, and intelligent systems. In parallel, scholarship on algorithmic management shows that digital systems increasingly mediate managerial control, intensify data collection, and redistribute authority between humans and machines (Keegan & Meijerink, 2025; Zhang et al., 2025). In HRM, these developments generate not only efficiency gains but also governance challenges related to surveillance, contestability, accountability, and employee trust (Gong et al., 2024; Zhang et al., 2025).

Despite growing research on responsible AI, digital transformation, and AI-enabled HRM, the literature remains fragmented in three important respects. First, digital business scholarship has not sufficiently theorized HRM as a strategic governance site within digital transformation. Second, HRM research has often examined AI applications and ethical concerns without fully connecting them to broader digital business outcomes such as legitimacy, trust, and sustainable value creation. Third, responsible AI governance research has generated useful principles and review frameworks, yet there is still limited integrative theorization explaining how human-centric AI governance in HRM can support responsible digital business more broadly (Bujold et al., 2024; Papagiannidis et al., 2025; Ritter & Pedersen, 2020). This fragmentation creates a clear opportunity for a conceptual contribution that bridges digital business, AI governance, and HRM.

Accordingly, this paper develops a human-centric conceptual framework for AI governance in digital HRM and positions it as a foundation for responsible digital business. The paper argues that AI governance in HRM should be understood as a sociotechnical and strategic capability rather than a compliance add-on. A human-centric approach is especially important because HRM is where algorithmic systems most visibly affect fairness, dignity, voice, and trust (Bujold et al., 2024; Papagiannidis et al., 2025). By integrating insights from digital transformation, responsible AI governance, algorithmic HRM, and recent future-of-work scholarship, the paper explains how governance architecture, human-centric design principles, and digital HRM application domains can jointly shape responsible digital business outcomes.

This paper makes three contributions. First, it extends digital business scholarship by conceptualizing digital HRM as a strategically important site of AI governance within broader digital transformation (Digital Business, n.d.; Ritter & Pedersen, 2020). Second, it contributes to HRM and responsible AI literature by moving beyond isolated ethical concerns toward an integrated governance framework (Bujold et al., 2024; Kim et al., 2025). Third, it offers a foundation for future empirical research on how organizations can govern AI-enabled people management in ways that support both business performance and human-centered responsibility (Papagiannidis et al., 2025; Úbeda-García et al., 2025). The remainder of the paper situates digital HRM within digital business transformation, clarifies the conceptual foundations and theoretical anchoring of the framework, identifies the key governance challenges in AI-driven HRM, and then presents the proposed framework and its implications for responsible digital business.

2. Digital HRM as a Domain of Digital Business Transformation

Digital HRM should be understood not merely as the technological modernization of HR processes, but as part of the broader digital transformation of the firm. Digital transformation reshapes organizational structures, managerial capabilities, value creation logic, and the future of work rather than simply adding new tools to existing routines (Vial, 2019; Ritter & Pedersen, 2020). Within this perspective, digital HRM becomes more than an administrative support function; it becomes a strategic component of digital business transformation.

The digital transformation literature emphasizes that digitalization alters how firms create and capture value by reconfiguring capabilities, relationships, and organizational logic (Ritter & Pedersen, 2020; Vial, 2019). HRM is central to this process because it helps shape the human and organizational conditions through which digital strategies are implemented and sustained. Elia et al. (2024) argue that successful digital transformation depends on aligning technological initiatives with roles, competencies, behaviors, and organizational enablers. This insight is directly relevant to digital HRM, because HR systems influence whether organizations develop the skills, leadership practices, and governance routines needed for responsible digital transformation.

From this perspective, digital HRM can be defined as the use of digital and intelligent technologies to support, augment, and transform HR activities in ways that increase automation, data intensity, integration, and strategic responsiveness. While earlier forms of e-HRM focused mainly on administrative efficiency and information accessibility, digital HRM increasingly involves analytics, platform-based infrastructures, AI-supported decision systems, and algorithmic forms of people management (Tambe et al., 2019; Gong et al., 2024). Recent work further suggests that digital HRM is increasingly being viewed as an important organizational pathway for achieving more integrated and sustainable management practices, reinforcing its strategic relevance within broader digital transformation efforts (Virmani et al., 2025). This shift matters because HRM is no longer simply using technology; it is increasingly being reorganized by digital logic.

AI has accelerated this development. Contemporary HR systems increasingly incorporate AI in recruitment, selection, performance management, workforce planning, learning personalization, and employee-service platforms (Tambe et al., 2019; Úbeda-García et al., 2025). As these technologies expand, HR decisions become more datafied, predictive, and scalable. At the same time, HRM becomes a key site where organizations confront the consequences of algorithmic decision-making, including new questions about control, fairness, and accountability (Bujold et al., 2024; Kim et al., 2025).

This is why digital HRM should be viewed as a domain of digital business rather than a purely functional HR innovation. AI-enabled HRM affects how organizations identify talent, evaluate performance, allocate opportunities, and govern employee behavior. These changes are strategically significant because they shape trust, legitimacy, inclusion, productivity, and the social sustainability of the enterprise. In this sense, digital HRM is both an object of digital transformation and a mechanism through which broader digital business transformation is enacted (Elia et al., 2024).

This interpretation is reinforced by emerging research on algorithmic HRM. Gong et al. (2024) show that algorithmic HRM should be understood not only through technical features but also through the affordances it creates for awareness, control, optimization, and coordination. Thus, algorithmic systems do not merely automate HR tasks; they actively reshape how HR activities are interpreted, organized, and governed. These affordances may enhance efficiency and decision coordination, but they may also intensify surveillance, reduce contestability, and displace human judgment if not governed appropriately (Gong et al., 2024; Zhang et al., 2025).

The future-of-work dimension further strengthens the relevance of digital HRM to digital business. As algorithmic systems increasingly shape how work is monitored, organized, and evaluated, HRM becomes central to the governance of digital work (Kim et al., 2025; Zhang et al., 2025). This expanded role means that digital HRM is no longer limited to staffing and development; it also helps determine how organizations balance efficiency, control, employee experience, and legitimacy in digitally mediated workplaces.

For this reason, digital HRM is strategically consequential for responsible digital business. Responsible digital business requires more than technological adoption; it requires organizational arrangements that align innovation with fairness, accountability, human dignity, and sustainable value creation (Papagiannidis et al., 2025). Because HRM governs access to work, assessment, development, and workplace experience, it is one of the most visible arenas in which employees and applicants encounter the ethical and practical consequences of AI. Failures in digital HRM can therefore damage not only employee outcomes but also organizational trust and legitimacy. Conversely, well-governed digital HRM can help translate digital transformation into responsible business conduct.

Taken together, the literature suggests that digital HRM should be positioned as a strategic domain of digital business transformation because it is increasingly shaped by AI, influences the organizational conditions of digital transformation, and mediates how the future of work is governed (Elia et al., 2024; Kim et al., 2025; Úbeda-García et al., 2025; Zhang et al., 2025). Any serious conceptual account of responsible digital business should therefore treat AI-enabled HRM not as a peripheral subtopic, but as a central governance domain within digital transformation.

This interpretation provides the basis for the next section, which clarifies the conceptual foundations of the paper by defining human-centric AI governance, digital HRM, responsible digital business, and algorithmic decision-making in HRM.

3. Conceptual Foundations

A strong conceptual paper requires construct clarity. In this study, four concepts are central: human-centric AI governance, AI governance, digital HRM, and responsible digital business. Clarifying these constructs is important because the literature often treats them in overlapping but insufficiently integrated ways, limiting cumulative theory building in the fields of responsible AI, digital transformation, and algorithmic HRM (Papagiannidis et al., 2025; Vial, 2019; Gong et al., 2024). This section therefore defines the core constructs used in the paper and establishes the conceptual boundaries of the proposed framework.

3.1 Human-Centric AI Governance

Human-centric AI governance refers to the principles, structures, and practices through which organizations ensure that AI systems remain aligned with human values, human oversight, and human wellbeing throughout their design, deployment, and use. The concept draws on the broader human-centered AI movement, which argues that AI should augment rather than displace human agency and should be designed to remain reliable, controllable, and responsive to human needs (Shneiderman, 2022). Recent organizational research similarly suggests that human-centered AI is especially important when AI systems affect employees' autonomy, perceptions, and opportunities because such systems can redistribute control in ways that are not always visible to those affected (Dong et al., 2024; Kluge et al., 2024).

In this paper, human-centricity extends beyond interface design or usability. It refers to a broader normative and organizational orientation in which AI governance protects dignity, supports contestability, preserves meaningful human judgment, and takes seriously the lived experience of those subject to algorithmic decisions. This interpretation is consistent with recent discussions of human-centric AI governance that emphasize alignment with human values and the public good rather than narrow technical performance criteria (Wittmann & Meynhardt, 2025). Applied to HRM, human-centric governance means that AI should not only improve efficiency, but also protect employee voice, fairness, privacy, and access to human review in consequential employment decisions.

3.2 AI Governance

AI governance can be defined as the set of formal and informal mechanisms through which organizations direct, monitor, evaluate, and control the development and use of AI systems. It includes policies, accountability structures, standards, audit processes, oversight routines, and procedures for identifying and addressing risks. Papagiannidis et al. (2025) conceptualize responsible AI governance through structural, relational, and procedural practices, arguing that effective governance spans the entire AI lifecycle rather than being confined to ex post compliance. This is important because it frames governance as an organizational capability rather than a narrow legal or technical safeguard.

This interpretation is especially relevant in HRM, where AI systems often involve external vendors, internal data teams, HR professionals, and line managers simultaneously. In such contexts, governance failure may stem less from the absence of ethical principles than from fragmented responsibility, weak oversight, or poor procedural integration. AI governance in this paper therefore refers not only to organizational intentions, but also to the arrangements through which responsibility is allocated and exercised in practice (Papagiannidis et al., 2025; de Almeida et al., 2025). This interpretation is reinforced by recent review evidence showing that responsible AI in real-world settings depends on a combination of governance practices spanning accountability, oversight, transparency, stakeholder involvement, and ongoing monitoring rather than on isolated principles alone (Bach et al., 2025).

3.3 Digital HRM

Digital HRM refers to the use of digital and intelligent technologies to support, augment, or transform HR activities and employee-related decision processes. The concept moves beyond earlier notions of e-HRM, which focused primarily on administrative digitization and information accessibility, toward a more advanced configuration in which analytics, AI, platform infrastructures, and algorithmic systems are embedded in core HR functions (Tambe et al., 2019; Gong et al., 2024). In this sense, digital HRM is not simply HR with software; it is HRM increasingly shaped by datafication, automation, predictive logic, and digital coordination.

Recent research confirms that AI-enabled HRM now spans recruitment, selection, performance management, workforce planning, employee-service systems, and development activities (Bujold et al., 2024; Úbeda-García et al., 2025). Gong et al. (2024) further argue that algorithmic HRM should be understood through the functional affordances it creates, including visibility, optimization, coordination, and control. Accordingly, this paper treats digital HRM as a strategic and sociotechnical domain of digital business transformation rather than merely an operational support system.

3.4 Responsible Digital Business

Responsible digital business refers to digitally enabled value creation that remains aligned with ethical, social, and governance expectations. It involves the pursuit of innovation and efficiency through digital technologies without undermining trust, fairness, legitimacy, or human wellbeing. Although the exact term is less established than concepts such as responsible AI or digital transformation, the underlying logic is supported by adjacent scholarship showing that digital transformation reshapes organizational value creation while governance determines whether digital innovation remains legitimate and socially acceptable (Ritter & Pedersen, 2020; Vial, 2019; Papagiannidis et al., 2025).

For the purposes of this paper, responsible digital business is defined as an organizational condition in which digital transformation is pursued through governance arrangements that protect stakeholders and sustain legitimate value creation over time. This concept is especially relevant here because AI in HRM affects both organizational performance and stakeholder experience. When AI-enabled HRM is poorly governed, organizations may gain efficiency while eroding legitimacy, trust, and ethical resilience. When governed in human-centric ways, however, digital HRM can support both productivity and socially sustainable business outcomes.

3.5 Algorithmic Decision-Making in HRM

Algorithmic decision-making in HRM refers to the use of computational systems to support, recommend, rank, predict, or automate decisions related to employees and applicants. These systems may operate fully automatically or in hybrid form, where human decision-makers remain formally responsible but rely heavily on algorithmic outputs. Research on algorithmic management shows that such systems increasingly influence work allocation, evaluation, monitoring, and managerial control, thereby reshaping both the practical and symbolic foundations of authority at work (Zhang et al., 2025). In HRM, this includes candidate screening, interview scoring, promotion recommendations, attrition prediction, performance classification, and productivity assessment (Tambe et al., 2019; Gong et al., 2024). Recent work further suggests that algorithmic decision-making in HRM should be evaluated against principles of appropriateness rather than only acceptance or aversion, particularly in contexts involving consequential judgments about people (Strohmeier et al., 2025).

This concept is central because the governance challenges addressed in this paper become most visible when algorithmic systems shape consequential employment outcomes. Algorithmic decision-making is problematic not simply because it uses data or automation, but because it can obscure responsibility, reduce contestability, and amplify biases while presenting decisions as objective or efficient. Bujold et al. (2024) therefore show that responsible AI in HRM must be assessed not only by technical capability, but also by fairness, transparency, and quality-of-work implications.

3.6 Conceptual Boundary of the Study

Taken together, these definitions clarify the paper's conceptual position. First, the study focuses specifically on AI governance in digital HRM, rather than AI across all organizational domains. Second, it treats governance as an organizational capability involving structures, relationships, and procedures rather than as a purely legal or regulatory matter (Papagiannidis et al., 2025). Third, it adopts a human-centric perspective that places human dignity, oversight, contestability, and stakeholder experience at the center of AI deployment in HRM (Shneiderman, 2022; Wittmann & Meynhardt, 2025). Finally, it connects these ideas to the broader goal of responsible digital business, thereby moving beyond a narrow HR ethics discussion toward a digital business framework concerned with legitimacy, trust, and sustainable value creation.

These conceptual foundations prepare the ground for the next section, which anchors the framework theoretically through sociotechnical systems theory, governance theory, and stakeholder theory.

4. Theoretical Anchoring of the Framework

A conceptual framework is stronger when grounded in a coherent theoretical logic rather than assembled as a descriptive list of issues. This paper anchors the proposed framework in three complementary perspectives: sociotechnical systems theory, governance theory, and stakeholder theory. This combination is appropriate because AI-enabled HRM operates as a digital business phenomenon in which technical systems, organizational arrangements, and stakeholder consequences are deeply intertwined.

4.1 Sociotechnical Systems Theory as the Primary Lens

Sociotechnical systems theory provides the primary lens because it emphasizes that organizational outcomes emerge from the interaction of social and technical elements rather than from technological efficiency alone. In digitally transformed organizations, AI systems are embedded within work practices, managerial routines, organizational structures, and human interpretations. This makes sociotechnical reasoning especially relevant to AI in HRM, where algorithmic tools shape consequential decisions about hiring, evaluation, development, and control.

Recent scholarship has reaffirmed the relevance of sociotechnical thinking in the digital era. Govers and van Amelsvoort (2023) argue that sociotechnical systems design remains highly relevant because digital technologies reshape the division of labor, work design, and coordination. Similarly, Colombari et al. (2024) show that digitalization affects organizational structures, decision processes, and skill requirements in interconnected ways rather than through a simple automation logic. These insights are directly relevant to digital HRM because AI changes not only the tools used by HR professionals, but also the relationships among data, managerial judgment, employee experience, and organizational control.

A sociotechnical lens helps avoid a purely technical reading of AI governance. A narrow view might reduce governance to model accuracy, data quality, or compliance. By contrast, sociotechnical reasoning suggests that the effectiveness and legitimacy of AI in HRM depend on how technical systems interact with human discretion, institutional norms, work design, and employee perceptions. This is consistent with Dong et al. (2024), who argue that AI in organizational settings should be studied in relation to human agency, managerial systems, and contextual design choices rather than as a stand-alone technology. From this perspective, human-centric AI governance must address both technological configuration and organizational design.

4.2 Governance Theory as a Supporting Lens

While sociotechnical systems theory explains the interdependence of social and technical elements, governance theory clarifies how responsibility, oversight, and coordination should be organized. This lens is necessary because the central issue in this paper is not merely the presence of AI in HRM, but whether organizations possess the structures and processes needed to direct and control its use responsibly.

Papagiannidis et al. (2025) conceptualize responsible AI governance through structural, relational, and procedural practices, arguing that governance must extend across the AI lifecycle and shape the conditions under which AI is developed, deployed, monitored, and revised. Likewise, de Almeida et al. (2025) show that implementing responsible AI governance requires formal structures, clear accountability, and operational mechanisms capable of translating principles into ongoing practice. Together, these contributions support the view that governance is not a symbolic label but an organizational capability.

This perspective is especially relevant in digital HRM, where AI systems often involve external vendors, internal analysts, HR specialists, and line managers simultaneously. In such settings, responsibility can easily become diffused unless roles, review mechanisms, and escalation procedures are clearly defined. Governance theory therefore helps explain why governance architecture appears as a central dimension in the proposed framework and why oversight, accountability, and review are necessary conditions for making AI-enabled HRM trustworthy and legitimate.

4.3 Stakeholder Theory as a Supporting Lens

The third theoretical anchor is stakeholder theory, which is important because AI governance in HRM has consequences for multiple parties, not only for organizational performance. AI systems in HRM affect applicants, employees, managers, executives, regulators, and broader societal expectations regarding fairness and non-discrimination.

A stakeholder perspective therefore helps explain why responsible digital business cannot assess AI in HRM solely in terms of efficiency or cost reduction.

Stakeholder scholarship emphasizes that business decisions should be evaluated in relation to value creation for, and impacts on, multiple stakeholder groups rather than shareholders alone. Dmytriyev et al. (2021) show that stakeholder theory is particularly useful for understanding managerial problems involving ethics, value, and organizational relationships. More recently, Valentinov (2025) highlights how stakeholder theory addresses enduring problems of business by clarifying how firms navigate competing claims while maintaining legitimacy. This is directly relevant here because AI in HRM creates trade-offs among efficiency, fairness, transparency, privacy, and employee dignity.

Within the proposed framework, stakeholder theory supports two moves. First, it broadens the evaluation of AI governance beyond internal managerial convenience. Second, it justifies the paper's focus on outcomes such as trust, legitimacy, inclusion, and sustainable value. A governance model that improves efficiency while harming key stakeholders would not be consistent with responsible digital business. Stakeholder theory therefore provides the normative and relational foundation for linking AI governance in HRM to broader organizational legitimacy.

4.4 Integrating the Three Lenses

Taken together, these three lenses offer a coherent foundation for the framework. Sociotechnical systems theory explains why AI in HRM must be understood as an interaction among technologies, people, structures, and work practices. Governance theory explains how organizations can structure accountability, oversight, and procedural control across that interaction. Stakeholder theory explains why the success of such governance must be judged not only by efficiency, but also by legitimacy, fairness, and value for affected groups.

This integration is particularly suitable for digital business research. Digital transformation is not simply a technical process; it is a reconfiguration of organizational value creation and work systems (Vial, 2019; Ritter & Pedersen, 2020). When AI enters HRM, this reconfiguration becomes especially visible because HRM sits at the boundary between business efficiency and human consequences. Recent HRM scholarship similarly shows that algorithmic technologies are reshaping the assumptions of strategic HRM and require new theorizing about the relationship between technology, management, and workforce outcomes (Kim et al., 2025). The combined use of sociotechnical systems theory, governance theory, and stakeholder theory therefore provides a strong basis for explaining why human-centric AI governance is necessary in digital HRM and how it can contribute to responsible digital business.

This theoretical anchoring also shapes the logic of the framework developed later in the paper. Specifically, it supports the argument that governance architecture influences the quality of AI use in HRM, that human-centric design principles shape stakeholder experience, and that responsible digital business outcomes emerge when technical capability is aligned with organizational oversight and stakeholder-sensitive value creation. The next section builds on this logic by identifying the governance challenges that make such a framework necessary.

5. Governance Challenges in AI-Driven Digital HRM

The need for a human-centric governance framework becomes clear when examining the challenges that accompany AI deployment in digital HRM. Although AI can improve speed, scale, and analytical sophistication in HR processes, its organizational use is frequently associated with governance gaps involving fairness, opacity, privacy, accountability, and trust (Bujold et al., 2024; Papagiannidis et al., 2025). These challenges are especially significant because HRM decisions affect access to employment, performance evaluation, development opportunities, and everyday work experience. As a result, governance weaknesses in AI-enabled HRM generate consequences that are simultaneously operational, ethical, and strategic.

5.1 Opacity and Limited Explainability

One of the most persistent governance challenges in AI-driven HRM is the opacity of algorithmic systems. Many AI tools used in recruitment, performance analytics, or employee assessment rely on models that are difficult for HR professionals, managers, and employees to interpret. This opacity weakens understanding of how decisions are made, what data are used, and how errors or biases can be identified and corrected (Papagiannidis et al., 2025). In HRM, where decisions affect hiring, promotion, appraisal, or termination, limited explainability becomes especially problematic.

Responsible AI scholarship suggests that transparency is not only a technical issue but also an organizational and relational one. Papagiannidis et al. (2025) emphasize that responsible AI governance depends on practices that make AI use understandable and reviewable across the lifecycle of deployment. In HRM, Bujold et al. (2024) similarly identify transparency and explainability as recurring conditions for responsible AI use. When applicants or employees cannot understand how an algorithmic system shaped a decision, organizations may struggle to justify outcomes, handle complaints, or preserve perceptions of procedural fairness. Opacity therefore undermines both accountability and legitimacy.

5.2 Bias, Discrimination, and Fairness Risks

A second challenge concerns bias and discrimination. AI systems in HRM are often trained on historical organizational data, labor-market data, or behavioral proxies that may reproduce past inequalities or encode new forms of disadvantage. This creates the risk that AI-assisted recruitment, screening, or performance assessment will generate systematically uneven outcomes across groups. Bujold et al. (2024) show that fairness concerns are among the most prominent themes in the responsible AI in HRM literature, especially in selection and evaluation contexts. Sony et al. (2025) similarly highlight the risk that AI-driven HRM systems may reinforce discrimination while appearing objective or neutral. Relatedly, Strohmeier et al. (2025) argue that appropriate algorithmic decision-making in HRM requires explicit attention to contextual fit, fairness, and the conditions under which automated judgments should remain open to human review.

These risks are not merely technical flaws; they are governance failures with legal, ethical, and reputational consequences. Recent reviews of algorithmic management further show that AI-driven managerial systems can intensify existing asymmetries of power and control if not designed and governed responsibly (Keegan & Meijerink, 2025; Zhang et al., 2025). In HRM, fairness cannot be treated as optional. It must be embedded in data selection, model design, oversight, review procedures, and appeal mechanisms. Without such safeguards, AI-enabled HRM may undermine both equal opportunity and organizational legitimacy.

5.3 Privacy, Surveillance, and Data Overreach

A third challenge concerns privacy and workplace surveillance. The digitalization of HRM has been accompanied by a significant expansion in the collection and analysis of employee-related data, including communication patterns, productivity metrics, behavioral signals, and engagement indicators. When AI is layered onto these systems, organizations gain new capacities for monitoring, prediction, and classification. Although such capabilities may support efficiency or workforce planning, they also raise questions about proportionality, consent, data minimization, and respect for employee dignity (Bujold et al., 2024).

The broader literature on algorithmic management suggests that AI often deepens managerial visibility over workers and intensifies forms of control that may be difficult to contest (Keegan & Meijerink, 2025; Zhang et al., 2025). This concern is also echoed in recent systematic review evidence showing that algorithmic management environments often intensify monitoring, reduce worker discretion, and heighten asymmetries of control, thereby reinforcing the governance relevance of privacy and surveillance concerns in digitally mediated work (Kadolkar et al., 2025). Recent OECD evidence similarly shows that algorithmic management is becoming a meaningful workplace governance issue for employers and employees, especially where monitoring, evaluation, and decision automation intersect. In HRM, this can shift people management toward continuous data extraction and behavioral optimization. From a governance perspective, the issue is not only what data can be collected, but what data should be collected and under what constraints. Privacy in this context is therefore not simply a compliance issue; it is a condition of maintaining trust, dignity, and acceptable boundaries in digital work systems (Milanez et al., 2025).

5.4 Accountability Diffusion

A fourth challenge is the diffusion of accountability across multiple actors and systems. AI in HRM is rarely designed and deployed by a single actor; it typically involves software vendors, data scientists, HR specialists, line managers, platform providers, and senior executives. In such arrangements, responsibility for outcomes can become fragmented. When a problematic decision occurs, it may be unclear whether responsibility lies with the model designer, the data source, the vendor, the HR function, or the manager who relied on the output. Papagiannidis et al. (2025) identify this diffusion of responsibility as a core governance problem in AI systems.

In HRM, accountability diffusion is especially serious because decisions often affect individuals in high-stakes and identity-relevant ways. Bujold et al. (2024) note that responsible AI in HRM requires not only ethical intentions but also identifiable structures of responsibility and review. Without clear accountability, organizations may rely on algorithmic outputs while denying meaningful ownership of the consequences. Human-centric governance therefore requires explicit allocation of roles, oversight responsibilities, and escalation pathways so that AI-supported HR decisions remain contestable and institutionally owned.

5.5 Erosion of Trust and Employee Voice

AI-driven HRM also creates governance challenges around trust and employee voice. HR decisions are often experienced as signals of how the organization values employees and applicants. When AI systems are used without transparency, consultation, or accessible appeal mechanisms, decisions may be perceived as impersonal, opaque, or unfair, thereby weakening trust in both the technology and the organization deploying it (Bujold et al., 2024). Recent evidence also shows that employee reactions to AI-supported career systems are shaped strongly by perceived fairness, underscoring the importance of trust and justice perceptions in AI-enabled people management (Köchling et al., 2025). Research on algorithmic management likewise suggests that AI-mediated control can reduce perceived autonomy and strain the social relationship between workers and organizations if governance arrangements do not preserve voice and meaningful human oversight (Keegan & Meijerink, 2025; Zhang et al., 2025).

Trust is strategically important because it supports acceptance of digital transformation, cooperation with organizational change, and perceptions of legitimacy. Once employees believe that decisions are governed by inaccessible systems they cannot question, resistance and cynicism may increase. Preserving trust therefore requires transparency, communication, reviewability, and procedural protections that signal the continuing relevance of human judgment and employee dignity.

5.6 Strategic Misalignment Between Efficiency and Human-Centric Values

A final challenge concerns the strategic misalignment that can arise when AI in HRM is optimized primarily for efficiency while neglecting broader organizational values. Digital business transformation often emphasizes speed, scalability, data-driven optimization, and competitive advantage. These aims are not inherently problematic, but in HRM they may conflict with fairness, inclusion, privacy, developmental support, and the relational dimensions of work. Kim et al. (2025) argue that algorithmic technologies are altering the foundations of strategic HRM and require new ways of balancing efficiency with managerial judgment and human consequences.

This tension is especially important because organizations may view AI-enabled HRM as a pathway to performance improvement without fully considering how governance failures can damage legitimacy, talent experience, or long-term trust. Recent responsible AI research in business and innovation further suggests that governance should be treated as an enabling capability rather than a barrier to innovation (Papagiannidis et al., 2025; Ye et al., 2025). The challenge, then, is not to oppose efficiency, but to ensure that efficiency is pursued within governance arrangements that preserve human-centric values. Without such alignment, digital HRM may produce short-term gains while undermining the social and ethical foundations of responsible digital business.

5.7 Implications for Framework Development

Taken together, these challenges show that AI in digital HRM cannot be governed adequately through isolated principles or ad hoc controls. Opacity weakens explainability, fairness risks threaten legitimacy, surveillance pressures erode dignity, accountability diffusion obscures responsibility, trust may deteriorate, and strategic optimization can detach digital HRM from human-centered values (Bujold et al., 2024; Papagiannidis et al., 2025; Zhang et al., 2025). These problems are interconnected rather than discrete. For example, limited explainability may intensify perceived unfairness, while weak accountability may amplify mistrust and reduce the credibility of human oversight.

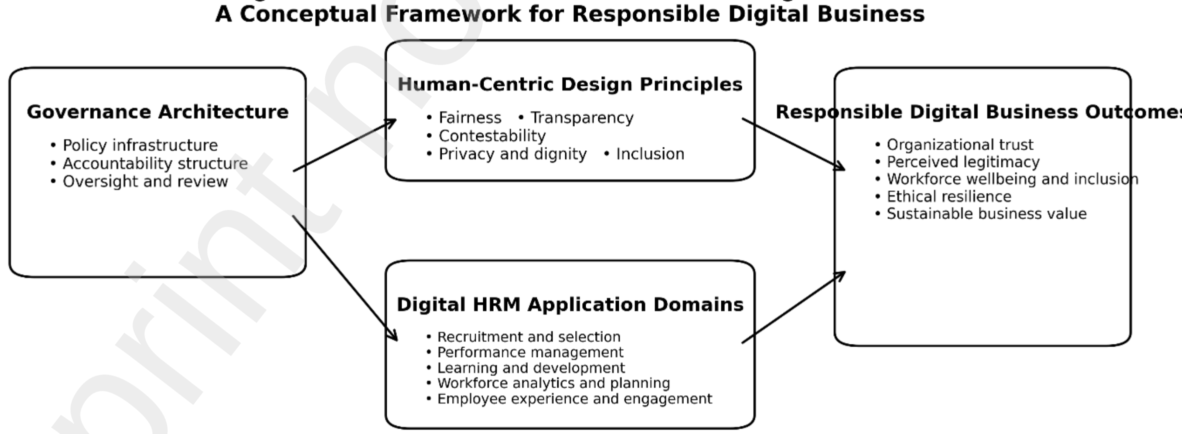
This interdependence creates the need for an integrated conceptual framework that links governance structures, human-centric design principles, application-specific risks, and broader responsible digital business outcomes. The next section responds to that need by presenting the proposed framework for human-centric AI governance in digital HRM.

6. Proposed Conceptual Framework

Building on the preceding discussion, this paper proposes a **human-centric conceptual framework for AI governance in digital HRM**. The framework explains how organizations can govern AI-enabled HRM in ways that support not only technical efficiency, but also fairness, legitimacy, trust, and sustainable digital business value. It responds directly to the governance challenges identified in the previous section and integrates the sociotechnical, governance, and stakeholder perspectives outlined earlier. At its core, the framework argues that responsible digital business outcomes in AI-enabled HRM depend on the alignment of four interrelated dimensions: **governance architecture**, **human-centric design principles**, **digital HRM application domains**, and **responsible digital business outcomes**.

This framework treats AI governance in HRM as a strategic organizational capability rather than a narrow compliance mechanism. This view is consistent with research showing that responsible AI governance must be institutionalized through structural, relational, and procedural practices across the AI lifecycle (Papagiannidis et al., 2025). It is also aligned with digital transformation scholarship, which emphasizes that digital technologies generate value only when embedded within coherent organizational arrangements, roles, and capabilities (Elia et al., 2024; Vial, 2019). In HRM, this means that responsible outcomes do not emerge automatically from AI adoption; they depend on how organizations structure governance, translate human-centric principles into practice, and manage the risks associated with specific HR domains.

**Figure 1. Human-Centric AI Governance in Digital HRM:
A Conceptual Framework for Responsible Digital Business**



The framework proposes that governance architecture shapes human-centric design principles
and the deployment of AI across digital HRM domains, which together influence responsible digital business outcomes.

Figure 1. Human-Centric AI Governance in Digital HRM: A Conceptual Framework for Responsible Digital Business

At a high level, the framework assumes that governance architecture provides the formal organizational basis for AI use in HRM, human-centric design principles shape how AI is configured and experienced, digital HRM application domains provide the contexts in which governance is enacted, and responsible digital business outcomes capture the strategic and relational consequences of well-governed AI deployment. Each dimension is explained below.

6.1 Governance Architecture

The first dimension, **governance architecture**, refers to the formal structures, processes, and accountability mechanisms through which AI use in HRM is directed, monitored, and reviewed. It is central because many AI-related problems in HRM arise not only from flawed models, but from weak organizational arrangements that fail to allocate responsibility clearly or embed oversight in routine practice (Papagiannidis et al., 2025; de Almeida et al., 2025).

This dimension includes three main components. The first is **policy infrastructure**, encompassing internal AI principles, acceptable-use standards, ethics guidelines, data governance rules, and decision protocols for HR applications. The second is **accountability structure**, which includes role clarity across HR leaders, line managers, analytics teams, executives, and external vendors. The third is **oversight and review**, including audit routines, escalation pathways, human review mechanisms, and periodic assessment of system performance and impact. Together, these elements form the institutional backbone of AI governance in HRM.

Within the framework, governance architecture serves two primary functions. First, it reduces accountability diffusion by clarifying who owns decisions, who monitors risks, and who can intervene when harmful outcomes emerge. Second, it increases organizational consistency by ensuring that AI use in HRM is not left to fragmented local experimentation. In this sense, governance architecture functions as the formal enabling condition for responsible digital business in people management.

6.2 Human-Centric Design Principles

The second dimension, **human-centric design principles**, translates governance intent into the practical and normative qualities of AI-enabled HRM systems. This dimension is essential because formal governance structures are insufficient unless they are reflected in how technologies are actually designed and used. Research on human-centered AI emphasizes that AI systems should preserve human agency, intelligibility, safety, and meaningful oversight rather than simply maximize automation or prediction accuracy (Shneiderman, 2022; Dong et al., 2024).

Five principles are especially important in the present framework. The first is **fairness**, meaning that AI systems should avoid unjust bias and support equitable treatment across individuals and groups. The second is **transparency**, referring to the degree to which AI-supported decisions can be understood, communicated, and scrutinized. The third is **contestability**, meaning that decisions influenced by AI should be open to questioning, review, or appeal. The fourth is **privacy and dignity**, which requires organizations to limit intrusive data practices and preserve acceptable boundaries in digitally mediated work. The fifth is **inclusion**, meaning that AI systems should not systematically exclude, disadvantage, or marginalize particular employee or applicant groups.

These principles matter because they shape how stakeholders experience AI in HRM. This is consistent with recent work showing that responsible AI principles generate meaningful outcomes only when enacted in complementary configurations rather than treated as isolated design features or symbolic commitments (Akbarighatar et al., 2026). A system may be technically accurate yet still be perceived as illegitimate if it is opaque, intrusive, or impossible to challenge. Conversely, human-centric design can strengthen trust and legitimacy by signaling that efficiency is being pursued without abandoning human values. In the framework, these principles therefore connect formal governance architecture to lived organizational experience.

6.3 Digital HRM Application Domains

The third dimension, **digital HRM application domains**, refers to the specific HR contexts in which AI systems are deployed. This dimension is included because governance needs vary depending on the nature of the HR activity, the sensitivity of the decision, and the type of data and consequences involved. Research on AI in HRM and algorithmic HRM shows that different HR domains involve distinct affordances, risks, and governance demands (Tambe et al., 2019; Gong et al., 2024).

The framework identifies five illustrative domains. The first is **talent acquisition and recruitment**, where AI may support sourcing, screening, ranking, and candidate assessment. The second is **performance management**, where AI may assist with productivity monitoring, evaluation, scoring, or behavioral analysis. The third is **learning and development**, where AI may shape personalized learning recommendations, adaptive pathways, or skill-gap analysis. The fourth is **workforce analytics and planning**, where predictive systems may support retention forecasting, succession planning, or labor allocation. The fifth is **employee experience and engagement**, where AI may be embedded in chatbots, sentiment analysis tools, wellbeing systems, or internal digital interaction platforms.

This dimension highlights that governance intensity is unlikely to be uniform across all HR domains. Recruitment and performance management, for example, may require stronger fairness, transparency, and contestability safeguards because they involve high-stakes judgments about opportunity and reward. By contrast, learning systems may place greater emphasis on transparency, inclusion, and developmental support. The application-domain dimension therefore introduces contextual variation while preserving the coherence of the overall framework.

6.4 Responsible Digital Business Outcomes

The fourth dimension, **responsible digital business outcomes**, captures the organizational and stakeholder consequences of well-governed AI deployment in digital HRM. This dimension is necessary because the purpose of governance is not limited to containing risk; it should also contribute positively to business legitimacy, stakeholder confidence, and sustainable value creation. This logic aligns with digital transformation research, which emphasizes that digital initiatives should be evaluated not only by efficiency but also by their long-term contribution to organizational capability and value (Ritter & Pedersen, 2020; Vial, 2019).

The framework identifies five outcomes. The first is **organizational trust**, referring to the confidence that employees, applicants, and managers place in AI-enabled HR systems and in the organization deploying them. The second is **perceived legitimacy**, meaning that AI use is viewed as appropriate, fair, and accountable. The third is **workforce wellbeing and inclusion**, capturing whether digital HRM supports humane and sustainable work practices rather than intensifying stress, exclusion, or surveillance. The fourth is **ethical resilience**, referring to the organization's ability to detect, absorb, and correct governance failures responsibly. The fifth is **sustainable business value**, meaning that digital transformation produces durable organizational benefits without eroding the relational and ethical foundations of the enterprise.

These outcomes move the framework beyond a narrow ethics vocabulary and position it more clearly within digital business scholarship. In this paper, responsible digital business is not treated as separate from digital performance. Rather, it is understood as the condition in which digital performance is sustained through governance arrangements that protect legitimacy, trust, and human-centered value over time.

6.5 Relationships Among the Four Dimensions

The framework proposes a directional logic linking the four dimensions. **Governance architecture** shapes the organizational conditions under which AI is introduced and controlled. **Human-centric design principles** determine whether those governance intentions are translated into system qualities that are visible and meaningful to stakeholders. **Digital HRM application domains** provide the contexts in which governance is enacted and where risk intensity varies. When these dimensions are aligned, organizations are more likely to achieve **responsible digital business outcomes**.

This logic can be summarized in three claims. First, governance architecture influences the quality and accountability of AI deployment in HRM by formalizing oversight, review, and responsibility. Second, human-centric design principles mediate how governance is experienced in practice, shaping trust, perceived fairness, and acceptability. Third, the effects of governance are conditioned by the characteristics of the HR domain in which AI is used, because some applications are more consequential, visible, and ethically sensitive than others. The combined result is the degree to which AI-enabled HRM contributes to or undermines responsible digital business.

6.6 Why the Framework Is Human-Centric

The framework is explicitly described as **human-centric** because it places affected people at the center of governance design and evaluation. This does not mean rejecting automation or digital efficiency. Rather, it means that AI systems in HRM should be judged partly by whether they preserve meaningful human oversight, protect dignity, allow challenge, and support sustainable work relationships (Shneiderman, 2022; Bujold et al., 2024). In HRM, where decisions shape employment access, career development, and organizational belonging, such a perspective is indispensable.

This emphasis also distinguishes the framework from approaches that treat governance primarily as risk compliance. Compliance remains important, but a human-centric model is broader: it asks whether AI systems support a form of digital business that remains acceptable to employees, attractive to talent, and legitimate in the eyes of stakeholders. In this sense, human-centric AI governance is not a constraint on digital transformation; it is a condition for making that transformation sustainable and socially defensible.

6.7 Conceptual Contribution of the Framework

The conceptual contribution of the framework lies in its integrative structure. Existing scholarship has generated important insights on responsible AI governance, algorithmic HRM, digital transformation, and the future of work, but these streams are often treated separately (Papagiannidis et al., 2025; Gong et al., 2024; Kim et al., 2025). The present framework brings them together by positioning digital HRM as a strategic digital business domain in which governance architecture and human-centric design jointly shape organizational legitimacy and value creation.

This integrative logic sets up the next section, which develops formal propositions from the framework. Those propositions specify how governance architecture, human-centric principles, and application-domain differences are expected to influence trust, legitimacy, and sustainable digital business outcomes in AI-enabled HRM.

7. Propositions Development

The framework developed in the previous section suggests that responsible digital business outcomes in AI-enabled HRM do not emerge automatically from technology adoption. Rather, they depend on the alignment of governance structures, human-centric design principles, and the characteristics of the HR domains in which AI is used. To make this logic explicit, this section derives a set of propositions from the framework. These propositions are not deterministic claims, but theoretically grounded expectations intended to guide future empirical research.

7.1 Governance Architecture and Responsible Digital Business Outcomes

The first proposition concerns **governance architecture**. Prior research shows that responsible AI governance depends on organizational practices that are structural, relational, and procedural rather than merely aspirational (Papagiannidis et al., 2025). AI systems are therefore more likely to generate responsible outcomes when organizations establish clear policies, identifiable accountability, review procedures, and oversight arrangements. In digital HRM, these features are especially important because AI-supported decisions often affect consequential employment outcomes and involve multiple actors, including vendors, data analysts, HR professionals, and line managers.

When governance architecture is weak, responsibility tends to fragment, oversight becomes inconsistent, and harmful outcomes are harder to detect and correct. By contrast, stronger governance architecture should increase the likelihood that AI systems in HRM are used in ways that are reviewable, accountable, and aligned with broader organizational values. This logic is consistent with research showing that digital transformation produces sustainable value only when digital initiatives are embedded within coherent organizational arrangements and capabilities (Vial, 2019). Accordingly:

P1. Stronger AI governance architecture in digital HRM is positively associated with responsible digital business outcomes, including organizational trust, perceived legitimacy, ethical resilience, and sustainable business value.

7.2 Human-Centric Design Principles and Organizational Trust

The second proposition concerns **human-centric design principles** as the mechanism through which governance becomes meaningful in practice. Human-centered AI research emphasizes that AI systems should preserve human agency, intelligibility, controllability, and alignment with human needs rather than merely optimize technical performance (Shneiderman, 2022; Dong et al., 2024). In HRM, this is especially important because employees and applicants do not encounter governance architecture directly; they encounter AI through the way systems are designed, explained, and experienced.

When AI-supported HR practices are fair, understandable, and open to review, stakeholders are more likely to perceive them as acceptable and trustworthy. By contrast, systems that are opaque, intrusive, or impossible to challenge are likely to weaken trust even if they improve efficiency. Empirical review work in HRM similarly confirms that transparency, fairness, and explainability are recurring conditions of responsible AI use in people management (Bujold et al., 2024). Thus:

P2. The stronger the human-centric design principles embedded in AI-enabled HRM systems, the higher the level of trust that employees, applicants, and managers place in those systems and in the organization deploying them.

7.3 Transparency and Contestability as Drivers of Perceived Fairness

A more specific proposition can be derived from the role of **transparency** and **contestability**. Transparency makes AI-supported decisions more understandable, while contestability ensures that such decisions remain open to questioning, appeal, or correction. These features are particularly important in HRM because employment-related decisions often shape opportunity, reward, and professional identity. Recent work on responsible AI governance emphasizes that explainability and reviewability are central to making AI accountable in organizational settings (Papagiannidis et al., 2025). Similarly, the HRM literature identifies transparency and fairness as recurring concerns in empirical studies of AI use in people management (Bujold et al., 2024). This is consistent with recent arguments that appropriate algorithmic decision-making in HRM depends not only on technical performance, but also on the transparency, contestability, and contextual suitability of algorithm-supported judgments (Strohmeier et al., 2025).

When decision processes are visible and challengeable, stakeholders are more likely to view them as procedurally fair. Conversely, when decisions are difficult to interpret or cannot be contested, perceptions of unfairness are likely to increase. Therefore:

P3. Greater transparency and contestability in AI-enabled HRM systems are positively associated with higher perceived fairness of algorithmic HR decisions.

7.4 Perceived Legitimacy as a Mediating Mechanism

The framework also suggests that **perceived legitimacy** plays a mediating role between governance inputs and broader business outcomes. A stakeholder-oriented view implies that the value of AI is determined not only by technical utility or cost savings, but also by whether key stakeholders regard its use as appropriate, fair, and accountable. In digital business contexts, legitimacy matters because it supports acceptance of innovation, organizational reputation, and long-term value creation. Research on responsible AI governance likewise suggests that trustworthy and responsible practices shape whether AI deployment is viewed as acceptable and sustainable in practice (Papagiannidis et al., 2025).

In HRM, legitimacy is especially important because AI directly affects individuals' opportunities and treatment. Governance architecture and human-centric principles may therefore influence digital business outcomes partly by shaping whether AI use is regarded as legitimate. If stakeholders perceive AI-enabled HRM as legitimate, organizations are more likely to maintain trust, reduce resistance, and sustain value from digital transformation. Accordingly:

P4. Perceived legitimacy mediates the relationship between human-centric AI governance in digital HRM and responsible digital business outcomes.

7.5 Application Domains as a Contingency Factor

The fifth proposition addresses **digital HRM application domains** as a contingency condition. AI is used across multiple HR activities, but the governance requirements of these activities are unlikely to be identical. Research on algorithmic HRM suggests that algorithmic systems create different functional affordances depending on the HR context in which they are used, including varying implications for awareness, optimization, coordination, and control (Gong et al., 2024). Likewise, strategic HRM scholarship on algorithmic technologies emphasizes that different applications raise distinct questions about judgment, decision sensitivity, and people-management consequences (Kim et al., 2025).

This suggests that the effectiveness of governance practices will vary across HR domains. Recruitment and performance management, for example, may require especially strong fairness, transparency, and contestability safeguards because they directly shape access to jobs, evaluations, and rewards. Learning systems or workforce planning tools may still require governance, but their risk profiles may differ. Thus:

P5. The positive effects of human-centric AI governance on responsible digital business outcomes vary across digital HRM application domains, with stronger effects in high-stakes domains such as recruitment and performance management.

7.6 Strategic Alignment and Sustainable Business Value

The final proposition concerns **strategic alignment**. Digital transformation research shows that digital technologies create value when they are aligned with broader organizational strategy, structures, and capabilities rather than adopted in isolation (Vial, 2019). Similarly, research on strategic HRM in the era of algorithmic technologies suggests that AI in people management must be connected to broader HR strategy and organizational purpose if it is to contribute meaningfully to performance and long-term value (Kim et al., 2025).

This is important because AI governance can be framed narrowly as risk mitigation or more broadly as a capability supporting responsible digital transformation. The latter view is more consistent with the present paper. When AI governance in HRM is aligned with digital business strategy, organizations should be better able to balance efficiency, fairness, trust, and sustainability. Therefore:

P6. Organizations that align AI governance in digital HRM with broader digital business strategy are more likely to achieve sustainable business value from AI-enabled HRM.

7.7 Overall Theoretical Logic of the Propositions

Taken together, the six propositions translate the framework into a coherent explanatory logic. Proposition 1 identifies governance architecture as the formal organizational foundation of responsible AI use in HRM. Propositions 2 and 3 emphasize that governance must be experienced through human-centric system qualities, especially trust-building, transparency, and contestability. Proposition 4 introduces legitimacy as a mechanism through which governance influences broader organizational outcomes. Proposition 5 recognizes that governance effects are unlikely to be uniform across HR domains. Proposition 6 connects AI governance in HRM to the strategic logic of digital business value creation.

This structure is important because it avoids reducing the framework to a normative checklist. Instead, it presents a theory-informed set of relationships that future research can test through surveys, case studies, mixed-method designs, or comparative analysis. The next section builds on these propositions by discussing the theoretical, managerial, and policy implications of the framework.

8. Discussion and Implications

This paper developed a **human-centric conceptual framework for AI governance in digital HRM** and positioned it as a foundation for **responsible digital business**. The central argument is that AI governance in HRM should not be treated as a secondary compliance concern or a narrow HR ethics issue. Rather, it should be understood as a strategic digital business challenge at the intersection of technological capability, organizational control, stakeholder legitimacy, and the future of work (Papagiannidis et al., 2025; Vial, 2019).

8.1 Theoretical Implications

The first contribution of the framework is theoretical. Research on digital transformation, responsible AI, and AI-enabled HRM has generated important insights, but these conversations often remain fragmented. Digital transformation studies have focused largely on organizational capabilities, business-model change, and technology-enabled value creation, often without examining HRM as a strategic governance site in its own right (Ritter & Pedersen, 2020; Vial, 2019). By contrast, HRM research has increasingly explored AI applications and algorithmic technologies, but often without fully connecting them to broader digital business outcomes such as legitimacy, trust, and sustainable value creation (Bujold et al., 2024; Kim et al., 2025). By integrating these streams, the present framework offers a more unified explanation of how AI in HRM matters for digital business.

A second contribution lies in positioning **digital HRM as a domain of digital business transformation**. Digital business scholarship often foregrounds customer-facing technologies, platform models, and data-driven innovation, while HRM is treated as secondary to strategic digital change. The framework challenges this assumption by showing that HRM is one of the organizational domains in which AI deployment is most consequential because it governs access to work, performance evaluation, developmental opportunity, and workplace experience. In this sense, AI-enabled HRM is not peripheral to digital business; it is one of the arenas in which digital transformation becomes materially visible and normatively contested. This argument is reinforced by recent work showing that algorithmic management has moved from an edge case to a central concern in organizational research and management practice (Keegan & Meijerink, 2025).

Third, the framework contributes to responsible AI scholarship by shifting attention from abstract principles toward an organizational and strategically grounded governance model. Recent business and management research shows that responsible AI has become a growing research domain concerned with governance, risk, innovation, and organizational legitimacy, rather than a purely ethical side issue (Dhaigude & Kamath, 2025; Papagiannidis et al., 2025). The present paper extends that insight by specifying how governance becomes meaningful in the domain of digital HRM and by linking governance architecture to outcomes such as trust, legitimacy, ethical resilience, and sustainable value. In this sense, the framework also responds to recent calls for more practice-oriented accounts of how AI principles are enacted and combined in real organizational settings (Akbarighatar et al., 2026).

Fourth, the framework contributes to HRM theory by reinforcing the need to understand AI-enabled HRM as a **sociotechnical phenomenon** rather than as a simple technological enhancement of HR processes. Research on algorithmic technologies in HRM suggests that these systems alter not only how decisions are made, but also how authority, visibility, and managerial judgment are configured (Gong et al., 2024; Kim et al., 2025). The present framework adds to this discussion by showing that governance arrangements and human-centric design principles are constitutive of AI's organizational meaning and consequences, not external to them.

8.2 Contribution to Digital Business Literature

The framework also contributes directly to digital business literature. A recurring theme in digital transformation research is that digital technologies create value only when aligned with organizational capabilities, structures, and strategic intent (Elia et al., 2024; Ritter & Pedersen, 2020). Yet business value generated through digital technologies may be difficult to sustain if governance arrangements are weak or if stakeholders perceive digital practices as unfair, opaque, or intrusive. This is especially relevant in AI-enabled HRM because the workforce is not merely an internal resource, but a central stakeholder group whose trust and acceptance shape the success of digital transformation.

By linking AI governance in HRM to responsible digital business outcomes, the paper broadens the digital business conversation in two ways. First, it highlights that digital business is not only about value creation, but also about the conditions under which digital value remains legitimate and socially sustainable. Second, it shows that HR systems are part of the digital business architecture of the firm, not merely downstream functions responding to technological change. This move is also consistent with recent research arguing that responsible AI can support responsible innovation when governance, stakeholder inclusion, and organizational capability are treated as mutually reinforcing rather than separate concerns (Ye et al., 2025).

8.3 Contribution to HRM and Responsible AI Literature

The framework also adds value to HRM and responsible AI scholarship by offering a more integrated explanation of how responsibility should be conceptualized in AI-enabled people management. Existing HRM studies often treat fairness, bias, privacy, and transparency separately, which is valuable but can fragment the understanding of governance. The present framework suggests that these concerns are interconnected and should be understood within a broader organizational system that combines governance architecture, human-centric design principles, domain-specific implementation, and stakeholder-oriented outcomes.

This perspective matters because responsible AI in HRM is often approached either through technical design concerns or through ethical warning narratives. Both are important, but neither is sufficient on its own. A technical focus may overlook the relational and organizational dimensions of AI use, while a purely ethical focus may not show how responsibility can be embedded in institutional structures and strategic processes. The proposed framework therefore offers a more cumulative basis for theorizing at the intersection of HRM and responsible AI (Bujold et al., 2024; Papagiannidis et al., 2025).

The framework also emphasizes that **human-centricity** should be treated as a central organizing logic rather than a rhetorical add-on. In many managerial discussions, human-centered AI is invoked in broad terms but not operationalized in relation to concrete decisions or organizational mechanisms. This is particularly important because recent evidence suggests that employee reactions to AI in career-related systems depend heavily on whether such systems are perceived as fair and procedurally just (Köchling et al., 2025). By specifying fairness, transparency, contestability, privacy, and oversight as core features of human-centric governance, the framework clarifies how human-centered AI principles can shape both HR governance and digital business outcomes.

8.4 Managerial Implications

The framework carries several managerial implications. First, managers should treat AI governance in HRM as a **strategic capability** rather than a technical afterthought. Organizations that adopt AI in HR processes without building policy infrastructure, accountability structures, and review mechanisms are likely to expose themselves to trust deficits, legitimacy problems, and avoidable governance failures (Papagiannidis et al., 2025). Senior leaders should therefore ensure that AI deployment in HRM is accompanied by clear governance ownership, cross-functional coordination, and institutional review processes.

Second, organizations should not evaluate AI in HRM solely in terms of efficiency gains or predictive accuracy. These metrics remain relevant, but they should be balanced against indicators such as perceived fairness, trust, reviewability, privacy protection, and user acceptance. This is particularly important in high-stakes domains such as recruitment and performance management, where poor governance can generate serious employee and reputational consequences.

Third, HR leaders should be positioned as central actors in digital governance rather than passive recipients of vendor technologies or data-science outputs. Because digital HRM sits at the intersection of technology, employment relations, and organizational culture, HR professionals should play a central role in defining acceptable use, safeguarding employee dignity, and ensuring that AI-supported decisions remain subject to human review. In practice, this may require governance committees, ethical review protocols, AI impact assessments, and clearer communication with employees about how digital HR systems are used.

Fourth, governance mechanisms should be calibrated to the sensitivity of the HR domain, the type of data involved, and the severity of potential harm. Recruitment, promotion, dismissal-related analytics, and performance assessment may require especially strong safeguards, whereas developmental applications may allow different balances between automation and discretion. A one-size-fits-all approach to AI governance in HRM is therefore unlikely to be effective.

8.5 Policy and Governance Implications

Beyond managerial practice, the framework carries policy implications for organizations and institutions concerned with responsible digital transformation. At the organizational level, it suggests the need for internal governance systems that formalize responsibility for AI in HRM. This includes role clarity, escalation mechanisms, audit routines, documentation standards, and avenues for contesting or reviewing AI-supported decisions. Such arrangements can improve both risk management and organizational legitimacy. Recent systematic review evidence likewise suggests that responsible AI practices are most effective when organizations institutionalize them as interconnected governance routines rather than treating them as stand-alone compliance measures (Bach et al., 2025).

At a broader institutional level, the framework supports the view that AI governance in employment contexts deserves particular scrutiny because employment-related decisions carry deep distributive and social consequences. Recent OECD evidence suggests that algorithmic management is already producing new governance questions for employers and policy actors, particularly in relation to monitoring, autonomy, and decision accountability in the workplace (Milanez et al., 2025). The framework does not propose legal rules, but it suggests that organizations should anticipate stronger expectations around explainability, fairness, reviewability, and stakeholder protection in AI-enabled people management.

8.6 Future Research Implications

The propositions developed in this paper open several directions for future empirical inquiry. Researchers could test the framework quantitatively by examining how governance architecture and human-centric design principles affect trust, legitimacy, and organizational outcomes across different HR domains. Comparative studies could explore whether governance effects vary across sectors, institutional environments, or levels of HR digital maturity. Qualitative case studies could examine how organizations negotiate tensions between efficiency and human-centric values when implementing AI in recruitment, performance management, or workforce analytics.

There is also potential for multilevel research. Future studies could investigate how governance arrangements at the organizational level shape employee perceptions of fairness, contestability, and dignity at the individual level. Other promising directions include examining how human-centric AI governance interacts with digital leadership, organizational culture, or strategic HRM capability, as well as how cross-national differences shape expectations around AI governance in HRM.

8.7 Concluding the Discussion

In sum, the discussion highlights that AI governance in digital HRM is not a peripheral ethics topic but a central issue for responsible digital business. The framework contributes theoretically by integrating digital business, governance, and HRM perspectives; practically by offering managers a structured way to think about responsible AI deployment in HRM; and prospectively by providing a basis for future empirical research. Its central message is that organizations are unlikely to achieve sustainable digital business outcomes from AI-enabled HRM unless governance arrangements preserve legitimacy, trust, and human-centered value alongside efficiency and innovation.

9. Conclusion

This paper developed a human-centric conceptual framework for AI governance in digital HRM and positioned it within the broader domain of responsible digital business. Its central argument is that AI in HRM should not be viewed merely as a functional innovation for improving efficiency, automation, or predictive accuracy. Rather, AI-enabled HRM should be understood as a strategically significant domain of digital business transformation in which value creation, governance capability, stakeholder legitimacy, and human consequences converge (Ritter & Pedersen, 2020; Vial, 2019).

The analysis showed that the growing integration of AI into recruitment, performance management, workforce analytics, learning, and employee interaction systems has made HRM one of the most consequential arenas of digital transformation. At the same time, this shift has exposed organizations to interrelated governance challenges, including opacity, fairness risks, privacy concerns, accountability diffusion, erosion of trust, and tension between efficiency and human-centered values (Bujold et al., 2024; Papagiannidis et al., 2025). These challenges indicate that responsible AI use in HRM cannot be secured through technical sophistication alone; it requires organizational arrangements that institutionalize oversight, accountability, transparency, contestability, and respect for human dignity.

In response, the paper proposed a framework built around four interconnected dimensions: governance architecture, human-centric design principles, digital HRM application domains, and responsible digital business outcomes. By integrating insights from digital transformation, responsible AI governance, and algorithmic HRM, the framework contributes to digital business scholarship by conceptualizing HRM as a strategic governance site within digital transformation. It also contributes to HRM scholarship by emphasizing that AI-enabled people management must be understood as a sociotechnical and governance-sensitive phenomenon rather than a purely technological advance (Gong et al., 2024; Kim et al., 2025).

The paper further developed a set of propositions to guide future empirical research on how governance architecture, human-centric design, legitimacy, application domains, and strategic alignment shape responsible digital business outcomes in AI-enabled HRM. These propositions provide a basis for future quantitative, qualitative, and mixed-method work on AI governance in people management.

From a practical standpoint, the paper underscores that organizations pursuing digital transformation should not separate AI adoption from governance design. If AI in HRM is implemented without accountability, reviewability, and stakeholder sensitivity, short-term efficiency gains may come at the cost of trust, legitimacy, and long-term sustainability. Conversely, when AI governance is designed in human-centric ways, digital HRM can support responsible digital business by aligning technological innovation with fairness, transparency, and socially sustainable value creation (Bujold et al., 2024; Shneiderman, 2022).

In sum, human-centric AI governance in digital HRM is not a peripheral ethical add-on, but a central condition for responsible digital business. As organizations continue to embed AI into core people-management processes, the future of digital business will depend not only on what intelligent systems can do, but also on how those systems are governed, experienced, and legitimized in practice.

Declaration on the use of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author used generative AI and AI-assisted technologies to assist with language improvement, text editing, and organizational refinement. All intellectual contributions, conceptual development, interpretation, and final decisions remain those of the author. The author has carefully reviewed and edited the output and accepts full responsibility for the content of this manuscript.

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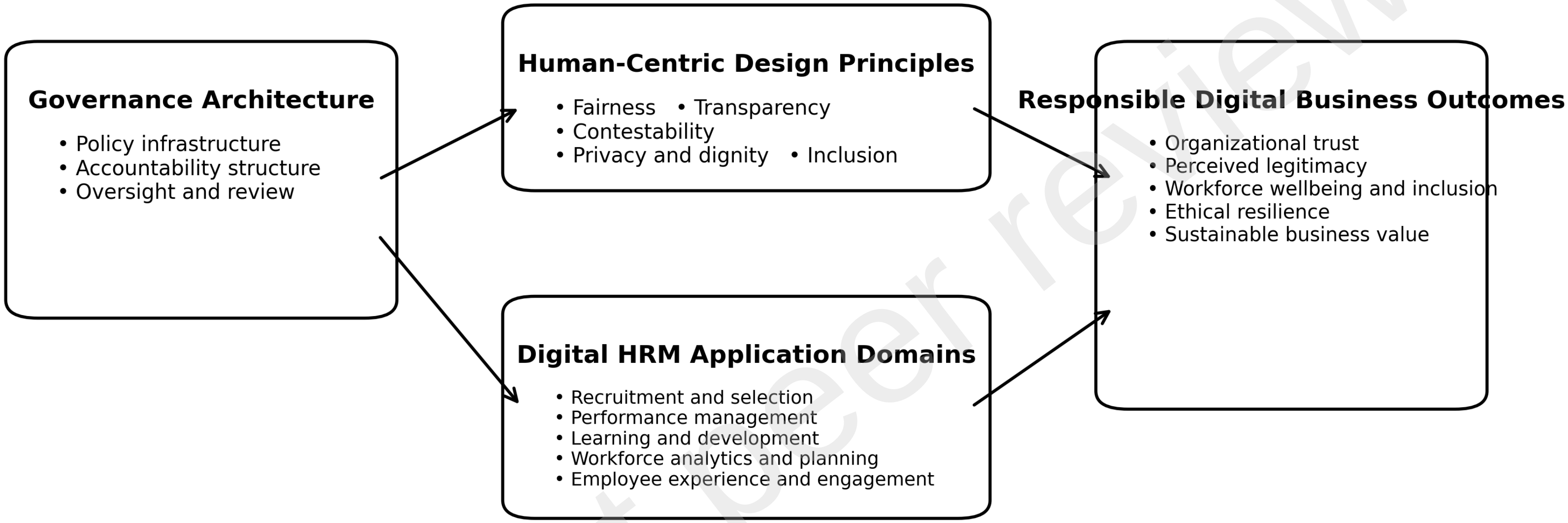
Figure 1. Human-Centric AI Governance in Digital HRM: A Conceptual Framework for Responsible Digital Business



The framework proposes that governance architecture shapes human-centric design principles and the deployment of AI across digital HRM domains, which together influence responsible digital business outcomes.